Thursday Reading Assessment: Chapter 7

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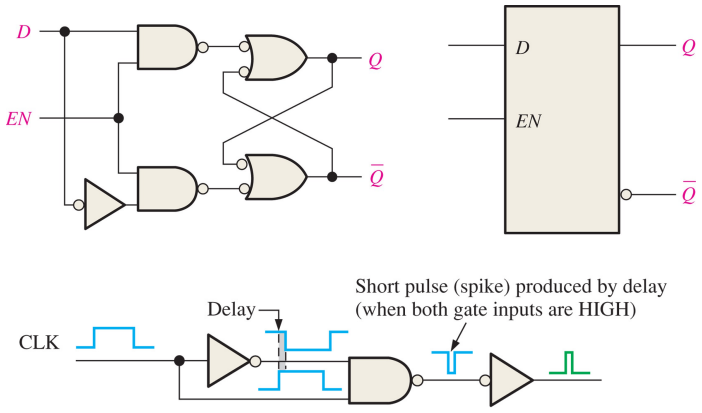


Figure 1: (Top) A gated D-latch system expressed in terms of NAND gates. (Bottom) A pulse transition detector.

1 Latches, Flip-Flops, and Timers: Gated D-latch

1. Produce the full truth table of the circuit in Fig. 1 (top) for the inputs EN and D, and the outputs Q and $\bar{Q}$. Remember to write NC for "No Change."

2. Create a timing diagram that enables the D-latch and passes the bitstream 10101010 from D to Q, and then disables the D-latch. What is the final state of the D-latch?

2 Latches, Flip-Flops, and Timers: Pulse Transition Detector

1. Consider the pulse transition detector (PTD) in Fig. 1 (bottom). Draw the diagram of a D flip-flop by comparing to the D-latch, replacing EN with the PTD.

2. If there is a 5 ns propagation delay in the initial inverter, and the initial CLK period is 1 μs, what is the final pulse duty cycle?

3. Consider the diagram of the JK flip-flop in Fig. 2. Assume $Q = 1$. What happens if $J = 1$ and $K = 1$?

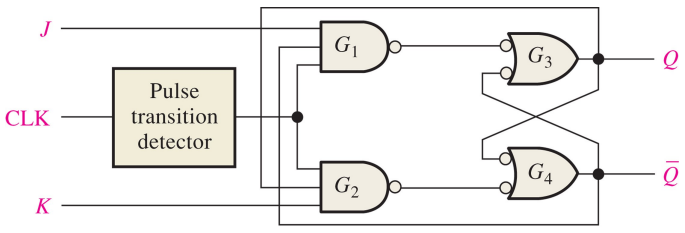


Figure 2: The layout of a JK flip-flop with PTD.